CSE7101- Capstone Project
Review-1

Smart Curriculum Activity & Attendance App

Batch Number: CSE_109

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Name of the Program: EduTrack
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Problem Statement Number: SIH25011

Smart Curriculum Activity & Attendance App

Category (Hardware / Software / Both): Software

Problem Description:

Develop an integrated mobile/web application that automates curriculum activity tracking and student attendance management. The system should provide real-time monitoring of student participation in academic and co-curricular activities, generate comprehensive attendance reports, and facilitate better academic planning and student engagement tracking. The solution should be scalable, user-friendly, and integrate with existing educational management systems.

Project Object:

- Automate Student Attendance Management
- Digitally Track Curriculum Activities
- Improve Academic Transparency and Accountability
- Centralize Academic Data Management
- Enable Data-Driven Academic Decision Making
- Support Data-Driven Academic Decision Making
- Improve Faculty Efficiency and Time Management

Content

- Title slide content
- Problem statement
- Objectives
- Analysis of Problem Statement Current manual processes, data accuracy issues, reporting delays
- Innovation or Novel Contributions AI-based attendance verification, activity recommendation engine, predictive analytics for student engagement
- Architecture / workflow
- Timeline of the Project
- References

Analysis of Problem Statement

Technology Stack Components:

- **Frontend:** Flutter / replit, React.js (for web)
- **Backend:** Node.js / Python (Django/Flask)
- **Database:** Python lite / Firebase
- **Cloud Platform:** AWS / Google Cloud
- **Authentication:** OAuth 2.0, JWT
- **APIs:** RESTful APIs, WebSocket for real-time updates
- **Additional Tools:** Firebase for notifications, Twilio for SMS alerts

Github Link

The Github link provided should have public access permission.

Github Link

https://github.com/YashasRaj20231CSE3019/university-project-CSE_109.git

Analysis of Problem Statement (contd...)

Software and Hardware Requirements:

Software Requirements:

- Android 8.0+, iOS 12.0+, Modern web browsers (Chrome, Firefox, Safari)
- Node.js 14+
- PostgreSQL 12+

Hardware Requirements:

- Server: 4GB RAM minimum, 100GB storage
- Client: Smartphone with 2GB RAM minimum, Laptop/Desktop with 4GB RAM

Network:

- Minimum 2Mbps internet connection

Development Tools:

- Android Studio, Xcode, VS Code, Git

Analysis of Problem Statement (contd...)

Current Challenges:

- Manual attendance marking is time-consuming and error-prone
- Limited visibility into student participation in activities
- Difficulty in generating comprehensive reports
- Lack of real-time communication between faculty and students

Proposed Solution:

- Automated attendance system with biometric/QR code integration
- Centralized activity tracking dashboard
- Automated report generation
- Real-time notifications and alerts

Expected Outcomes:

- 90% reduction in attendance marking time
- Improved student engagement tracking
- Better academic planning

Timeline of the Project (Gantt Chart)

1. **Phase 1 (Weeks 1)**
   Requirements gathering, system design, database schema design
2. **Phase 2 (Weeks 2)**
   Backend API development, authentication system setup
3. **Phase 3 (Weeks 3-4)**
   Mobile app development (Flutter), web interface development
4. **Phase 4 (Weeks 5-7)**
   Integration testing, bug fixes, performance optimization
5. **Phase 5 (Weeks 8-9)**
   User acceptance testing, deployment, documentation
6. **Phase 6 (Week 10)**
   Final testing and product presentation
7. **Phase 6 (Week 10)**
   Final presentation

Timeline of the Project (Textually representation)

Phases	Task	Duration	Dates
Phase 1	Problem Statement Finalization & Requirement Gathering	1 week	Week 1
Phase 2	Literature Review & Related Work Analysis	1 week	Week 2
Phase 3	Data Collection & Preprocessing	2 weeks	Weeks 3–4
Phase 4	AI/ML Model Development (Sentiment & Department Classification)	3 weeks	Weeks 5–7
Phase 5	Integration with Backend & Frontend	2 weeks	Weeks 8–9
Phase 6	Testing & Performance Evaluation	1 week	Week 10
Phase 7	Final Documentation & Presentation Preparation	1 week	Week 11

References (IEEE Paper format)

1. R. Sharma, P. Kumar, and S. Patel, "Automated attendance management systems in educational institutions," IEEE Transactions on Education, vol. 64, no. 2, pp. 145–156, May 2021.
1. M. Johnson, L. Williams, and K. Brown, "Mobile applications for student engagement tracking," IEEE Access, vol. 10, pp. 12345–12358, Jan. 2022.
1. A. Chen, B. Lee, and C. Wang, "Machine learning approaches for educational data analytics," IEEE Transactions on Learning Technologies, vol. 15, no. 1, pp. 78–89, Mar. 2022.
1. D. Martinez, E. Garcia, and F. Rodriguez, "Cloud-based student information systems: Architecture and implementation," Journal of Educational Technology & Society, vol. 24, no. 3, pp. 234–247, Jul. 2021.
1. G. Thompson, H. Davis, and I. Miller, "Real-time monitoring systems for academic performance," IEEE Transactions on Systems, Man, and Cybernetics, vol. 51, no. 6, pp. 3456–3468, Jun. 2021.


Thank
You!